

Date: 2026-05-11

Request ID: 100048502

Microwave				
Element(s)	Ca, Cr, Cu, Fe, K, Na, Ni, Zn			
SOP#				
Acid Cocktail	9 mL HCl, 3 mL HNO3, 3 mL H2SO4			
Rack	7			
Vessel	PTFE			
Stir	90%			
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)
15	1200	180	65	110
15	1500	240	65	110
30	1500	240	65	110
sample(s)	weight (g)	volume (mL)		
200127815_1	1.0045	50		
200127815_2	1.0017	50		
200127815_3	1.0118	50		
200127816_1	1.5011	50		
200127816_2	1.5053	50		
200127816_3	1.5019	50		

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200127815_1	50/50	1.476	73.47	0.01
200127815_2	50/50	1.474	73.57	0.01
200127815_3	50/50	1.550	76.60	0.01

ca result: 74.55 ppm

ca result: 0.01 %

ca oxide factor:

ca oxide result: 0.00 %

ca lot: *Manufacturer: Assurance, 28-35CAY exp: 2026-10-30*

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm	%Cr
200127815_1	50/50	0.139	6.92	0.00
200127815_2	50/50	0.138	6.89	0.00
200127815_3	50/50	0.143	7.07	0.00

cr result: 6.96 ppm

cr result: 0.00 %

cr oxide factor:

cr oxide result: 0.00 %

cr lot: *Manufacturer: Ricca, 45120423 exp: 2027-05-31*

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200127815_1	50/50	0.731	36.39	0.00
200127815_2	50/50	0.733	36.59	0.00
200127815_3	50/50	0.746	36.86	0.00

cu result: 36.61 ppm

cu result: 0.00 %

cu oxide factor:

cu oxide result: 0.00 %

cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

Fe ICP analysis

sample	Dilution	conc. [mg/L]	Fe_ppm	%Fe
200127815_1	50/50	0.878	43.70	0.00
200127815_2	50/50	0.819	40.88	0.00
200127815_3	50/50	0.812	40.13	0.00

fe result: 41.57 ppm

fe result: 0.00 %

fe oxide factor:

fe oxide result: 0.00 %

fe lot: *Manufacturer: Assurance, 28-75FEY exp: 2026-07-30*

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm	%K
200127815_1	50/50	0.975	48.53	0.00
200127815_2	50/50	0.793	39.58	0.00
200127815_3	50/50	0.881	43.54	0.00

k result: 43.88 ppm

k result: 0.00 %

k oxide factor:

k oxide result: 0.00 %

k lot: *Manufacturer: Assurance, 28-6KY exp: 2026-08-30*

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm	%Na
200127815_1	25/4	0.668	207.81	0.02
200127815_2	25/4	0.674	210.27	0.02
200127815_3	25/4	0.625	193.03	0.02

na result: 203.71 ppm

na result: 0.02 %

na oxide factor:

na oxide result: 0.00 %

na lot: *Manufacturer: Assurance, 27-159NAY exp: 2026-08-30*

Ni ICP analysis

sample	Dilution	conc. [mg/L]	Ni_ppm	%Ni
200127815_1	50/50	0.462	23.00	0.00
200127815_2	50/50	0.448	22.36	0.00
200127815_3	50/50	0.503	24.86	0.00

ni result: 23.41 ppm

ni result: 0.00 %

ni oxide factor:

ni oxide result: 0.00 %

ni lot: *Manufacturer: Assurance, 28-130NIT exp: 2026-10-30*

Zn ICP analysis

sample	Dilution	conc. [mg/L]	Zn_ppm	%Zn
200127815_1	50/50	0.007	0.35	0.00
200127815_2	50/50	0.019	0.95	0.00
200127815_3	50/50	0.000	0.00	0.00

zn result: 0.43 ppm

zn result: 0.00 %

zn oxide factor:

zn oxide result: 0.00 %

zn lot: *Manufacturer: Assurance, 28-58ZNY exp: 2026-10-30*

Note(s):

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Ca ICP analysis

sample	Dilution	conc. [mg/L]	Ca_ppm	%Ca
200127816_1	50/50	2.066	68.82	0.01
200127816_2	50/50	2.049	68.06	0.01
200127816_3	50/50	2.061	68.61	0.01

ca result: 68.50 ppm

ca result: 0.01 %

ca oxide factor:

ca oxide result: 0.00 %

ca lot: *Manufacturer: Assurance, 28-35CAY exp: 2026-10-30*

Cr ICP analysis

sample	Dilution	conc. [mg/L]	Cr_ppm	%Cr
200127816_1	50/50	0.126	4.20	0.00
200127816_2	50/50	0.099	3.29	0.00
200127816_3	50/50	0.095	3.16	0.00

cr result: 3.55 ppm

cr result: 0.00 %

cr oxide factor:

cr oxide result: 0.00 %

cr lot: *Manufacturer: Ricca, 45120423 exp: 2027-05-31*

Cu ICP analysis

sample	Dilution	conc. [mg/L]	Cu_ppm	%Cu
200127816_1	50/50	1.981	65.98	0.01
200127816_2	50/50	1.942	64.51	0.01
200127816_3	50/50	1.945	64.75	0.01

cu result: 65.08 ppm

cu result: 0.01 %

cu oxide factor:

cu oxide result: 0.00 %

cu lot: *Manufacturer: Assurance, 28-83CUT exp: 2026-09-30*

Fe ICP analysis

sample	Dilution	conc. [mg/L]	Fe_ppm	%Fe
200127816_1	50/50	0.917	30.54	0.00
200127816_2	50/50	0.647	21.49	0.00
200127816_3	50/50	0.591	19.68	0.00

fe result: 23.90 ppm

fe result: 0.00 %

fe oxide factor:

fe oxide result: 0.00 %

fe lot: *Manufacturer: Assurance, 28-75FEY exp: 2026-07-30*

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm	%K
200127816_1	50/50	1.328	44.23	0.00
200127816_2	50/50	1.264	41.98	0.00
200127816_3	50/50	1.301	43.31	0.00

k result: 43.18 ppm

k result: 0.00 %

k oxide factor:

k oxide result: 0.00 %

k lot: *Manufacturer: Assurance, 28-6KY exp: 2026-08-30*

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm	%Na
200127816_1	25/4	0.727	151.35	0.02
200127816_2	25/4	0.708	146.98	0.01
200127816_3	25/4	0.739	153.76	0.02

na result: 150.70 ppm

na result: 0.02 %

na oxide factor:

na oxide result: 0.00 %

na lot: *Manufacturer: Assurance, 27-159NAY exp: 2026-08-30*

Ni ICP analysis

sample	Dilution	conc. [mg/L]	Ni_ppm	%Ni
200127816_1	50/50	1.512	50.36	0.01
200127816_2	50/50	1.349	44.81	0.00
200127816_3	50/50	1.305	43.44	0.00

ni result: 46.21 ppm

ni result: 0.00 %

ni oxide factor:

ni oxide result: 0.00 %

ni lot: *Manufacturer: Assurance, 28-130NIT exp: 2026-10-30*

Zn ICP analysis

sample	Dilution	conc. [mg/L]	Zn_ppm	%Zn
200127816_1	50/50	0.000	0.00	0.00
200127816_2	50/50	0.000	0.00	0.00
200127816_3	50/50	0.000	0.00	0.00

zn result: 0.00 ppm

zn result: 0.00 %

zn oxide factor:

zn oxide result: 0.00 %

zn lot: *Manufacturer: Assurance, 28-58ZNY exp: 2026-10-30*

Note(s):

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$ppm\ M+ = [conc.][volume][dilution]/[weight]$$

$$\%M+ = [conc.][volume][dilution]/([weight]*10,000)$$

$$\%MO = [oxide\ factor]*\%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1-(\%LOI)/100])$$

$$\%M+ \ [dried] = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \ [dried] = \%MO / ([1-(\%LOI)/100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS])/([weight])$$

$$\%Cr_2O_3 = 1.462*[\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$